Inferring Strong Gravitational Lens Parameters from Images using Amortized Bayesian Inference

— Project Report —
Simulation Based Inference

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1 Introduction

Strong gravitational lensing occurs when a massive foreground galaxy bends the light of a distant background source, producing arcs, multiple images, or complete Einstein rings. The observed image morphology encodes the projected mass distribution of the lens: the ring radius traces the total enclosed mass via the Einstein radius θ_E, the arc shapes trace the lens ellipticity, and subtle distortions trace the tidal field of neighbouring structures (external shear). Extracting these parameters from images is a core task in astrophysical applications ranging from measuring dark-matter substructure to time-delay cosmography [6].

Classical lens modelling fits a parametric model to each image with likelihood-based sampling (e.g. MCMC), which is accurate but slow—often hours per system—and must be repeated from scratch for every new image. Upcoming surveys (Euclid, LSST) are expected to discover 10^4–10^5 new lenses, making per-system MCMC impractical. *Amortized* Bayesian inference (ABI) offers an alternative: a neural network is trained once on simulated (parameter, image) pairs to approximate the posterior $p(\boldsymbol{\theta} \mid \mathbf{x})$, after which inference for any new image requires only a forward pass. We implement this approach with BayesFlow [1, 2], using the strong-lensing simulator `lenstronomy` [3] as the forward model. The code is publicly available at https://github.com/5an528/strong-lens-inference.

2 Problem Definition

The objective is to estimate the physical parameters of a strong gravitational lens system from a *single* noisy telescope image. Each system consists of a singular isothermal ellipsoid (SIE) lens [4] with external shear, magnifying an elliptical Sérsic source. The inferred parameter vector is

$$\boldsymbol{\theta} = (\theta_E,\ e_1,\ e_2,\ \gamma_1,\ \gamma_2,\ x_s,\ y_s,\ R_s) \quad [+\,\kappa], \tag{1}$$

where θ_E is the Einstein radius, (e_1, e_2) the lens ellipticity components, (γ_1, γ_2) the external shear components, (x_s, y_s) the source position, R_s the source half-light radius, and κ an optional uniform external convergence (a "mass sheet"). Two questions structure the analysis:

1. How well can each parameter be recovered from one noisy image, and are the inferred uncertainties statistically calibrated?
2. When κ is added as a ninth parameter, does the inference correctly reflect the classical *mass-sheet degeneracy* [5, 6]—i.e. report that κ is essentially unidentifiable from a single image rather than hallucinating a tight constraint?

The challenge is that the likelihood $p(\mathbf{x} \mid \boldsymbol{\theta})$ of a 64×64-pixel image under PSF convolution and mixed Gaussian–Poisson noise is not available in closed form as a tractable function of $\boldsymbol{\theta}$, which motivates the simulation-based, likelihood-free approach.

3 Statistical model

3.1 Parameter Priors

Parameters are drawn from independent, physically motivated uniform priors (Table 1). Ellipticity is not sampled directly in (e_1, e_2): instead the axis ratio q and position angle ϕ are drawn and converted via $e = (1-q)/(1+q)$, $e_1 = e\cos 2\phi$, $e_2 = e \sin 2\phi$ (the convention of `lenstronomy` [3]), which guarantees every draw is a valid ellipse ($e_1^2 + e_2^2 < 1$). The external shear is likewise drawn in modulus and angle and converted to (γ_1, γ_2). This induces ring-shaped joint supports in the Cartesian components, visible in the prior pairplot (Figure 7 in the appendix). The ranges are chosen so that the lensed images remain informative: $\theta_E \in (0.7'', 1.6'')$

keeps the Einstein ring inside the field of view; $q \geq 0.6$ and $\gamma_{\text{ext}} \leq 0.08$ cover the typical ranges observed for galaxy-scale lenses [4, 6]; the source offset and size ranges guarantee visibly lensed (arc/ring) configurations rather than unlensed point sources. Identical priors are used at train and test time, a prerequisite for the calibration diagnostics of Section 6.2 to be valid.

Table 1: Prior ranges. Lengths in arcsec, angles in radians. All priors are uniform.

Parameter	Range	Meaning
θ_E	$(0.7,\ 1.6)$	Einstein radius
q	$(0.6,\ 1.0)$	lens axis ratio $\to (e_1, e_2)$
ϕ	$(0,\ \pi)$	lens position angle
γ_{ext}	$(0,\ 0.08)$	external shear modulus $\to (\gamma_1, \gamma_2)$
ϕ_{ext}	$(0,\ \pi)$	external shear angle
$x_s,\ y_s$	$(-0.3,\ 0.3)$	source position
R_s	$(0.05,\ 0.25)$	source half-light radius
κ	$(0,\ 0.20)$	external convergence (9-parameter run only)

Nuisance quantities are fixed and assumed known, following the assignment specification: the lens centroid (measurable from the lens light), the source amplitude, the source Sérsic index ($n = 2$), and the source-light ellipticity.

3.2 Likelihood

The data $\mathbf{x}$ is a 64×64 image. Writing the noise-free, PSF-convolved lenstronomy render as $\mu(\boldsymbol{\theta})$, each pixel i is corrupted independently by Gaussian background (sky/read) noise and Poisson shot noise:

$$x_i \mid \boldsymbol{\theta} \ \sim\ \mathcal{N}\Big(\mu_i(\boldsymbol{\theta}),\ \sigma^2_{\text{bkg}} + \mu_i(\boldsymbol{\theta})/t_{\text{exp}}\Big), \qquad i = 1, \ldots, 64^2, \tag{2}$$

with fixed $\sigma_{\text{bkg}} = 0.1$ and effective exposure time $t_{\text{exp}} = 100$. Unlike an idealized homoscedastic model, the noise variance grows with the signal itself (brighter arc pixels are noisier in absolute terms), matching real CCD behaviour; both components are applied with `lenstronomy`'s own noise utilities. Although the pixels are conditionally independent given $\boldsymbol{\theta}$, they are far from exchangeable—the information about $\boldsymbol{\theta}$ lies in the spatial *structure* of the image—which dictates a convolutional summary network (Section 5) rather than a permutation-invariant one.

The image dimensions were chosen deliberately: 64 pixels at $0.05''$/pixel give a $3.2''$ field of view, comfortably containing the largest Einstein ring allowed by the prior ($2\theta_E^{\max} = 3.2''$) while keeping the PSF (Gaussian, FWHM $0.1'' = 2$ pixels) resolved at Nyquist sampling. Larger grids make every simulation and every CNN forward pass proportionally slower without adding usable information at this PSF width.

4 Data Generation

Synthetic datasets were generated with the Python package `lenstronomy` [3], a widely used multi-purpose gravitational-lensing code: for each prior draw the SIE + shear (+ convergence) deflection field lenses the Sérsic source, the image is PSF-convolved, pixelated, and degraded with the two-component noise model above. Sampling from the priors was realized with `numpy` [9].

The source amplitude sets the signal-to-noise ratio and was tuned by a sweep over prior draws to place typical images in the realistic peak-SNR range of 10–30 targeted by the assignment (Table 2). With the selected amplitude of 150, the realized peak SNR of the final training set is median 16.1 (range 4.6–27.4). An earlier iteration of this project used the untuned amplitude of 20 (median peak SNR ≈ 4); it is retained in Section 6.5 as a controlled comparison of how noise destroys parameter information.

Table 2: Peak-SNR sweep over 60 prior draws used to tune the source amplitude into the assignment's target range of 10–30.

Source amplitude	min	median	max
20 (initial)	1.0	3.8	5.5
60	2.7	8.9	12.1
100	4.1	12.6	16.7
150 (selected)	5.7	**16.3**	21.3
200	7.1	19.4	25.1

Because `lenstronomy` simulation is CPU-bound and is the slowest step of the pipeline, the dataset is precomputed *offline*, parallelized over CPU worker processes, and stored; training then never touches the simulator. We generate $N_{\text{train}} = 50{,}000$ training, $N_{\text{val}} = 2{,}000$ validation, and $N_{\text{test}} = 300$ held-out test systems—at the upper end of the assignment's suggested budget of 10^4–10^5 simulations. This size was reached in steps: 8,000 samples produced clear overfitting, 20,000 removed it, and 50,000 further improved the recovery of the weakly identified shear and ellipticity parameters (Section 6.5). Example images and the prior pairplot are shown in Figures 8 and 7 in the appendix.

5 Neural Posterior Estimation and Training

BayesFlow [1, 2], a Python framework for amortized Bayesian inference built on Keras 3, was used to train a neural estimator of the posterior,

$$\hat{p}\big(\boldsymbol{\theta} \mid h(\mathbf{x})\big) \;\approx\; p(\boldsymbol{\theta} \mid \mathbf{x}), \tag{3}$$

where $h(\mathbf{x})$ is a learned summary statistic. Two networks are trained jointly end-to-end (Table 3): a **summary network** that compresses each $64 \times 64 \times 1$ image into a 48-dimensional feature vector, and an **inference network** that approximates the posterior given that summary. Amortization means the posterior is learned as a *function* of the data rather than approximated per observation: training is slower, but inference on any new image afterwards takes milliseconds.

Because the data are images, BayesFlow's built-in summary networks (designed for exchangeable sets and time series) do not apply, and a custom convolutional network is supplied: three convolutional blocks (two 3×3 convolutions each with 32, 64, and 128 filters, max-pooling, batch normalization), global average pooling, and two dense layers. The convolutional structure matches the spatial dependence of the data noted in Section 3.2, capturing the ring/arc morphology at successively coarser scales. As the inference network we use a conditional coupling-flow normalizing flow of depth 4; the physical parameters are standardized (z-scored) before entering the flow, since they span two orders of magnitude in scale ($\theta_E \sim 1$ vs. $\gamma_i \sim 10^{-2}$).

Table 3: Model architecture.

Parameter	Value
Summary network	custom CNN
Conv blocks	3 (32, 64, 128 filters)
Kernel size	3×3
Pooling	max + global average
Summary dim.	48
Activation	ReLU
Inference network	CouplingFlow
Depth	4
Standardization	inference variables

Table 4: Training configuration.

Parameter	Value
Training samples	50 000
Validation samples	2 000
Test samples	300
Epochs (max)	40
Batch size	32
Optimizer	Adam
Learning rate	10^{-3}
Early stopping	patience 10 (val. loss)
Backend	Keras 3 / PyTorch (CUDA)

Training is offline on the precomputed dataset (Table 4). Early stopping on the validation loss (patience 10, best weights restored) guards against the late-epoch overfitting observed in the small-dataset pilot runs; at 50,000 samples it never triggers, because the validation loss decreases essentially monotonically to the end (Figure 1), finishing at -8.05 with no train–validation gap. On an NVIDIA GeForce RTX 3060 the 8-parameter model trains in 123 minutes (the same configuration needed ≈ 4–$5\times$ longer on CPU, which is what made the larger dataset practical); the 9-parameter variant behaves identically (123 minutes, final validation loss -6.79).

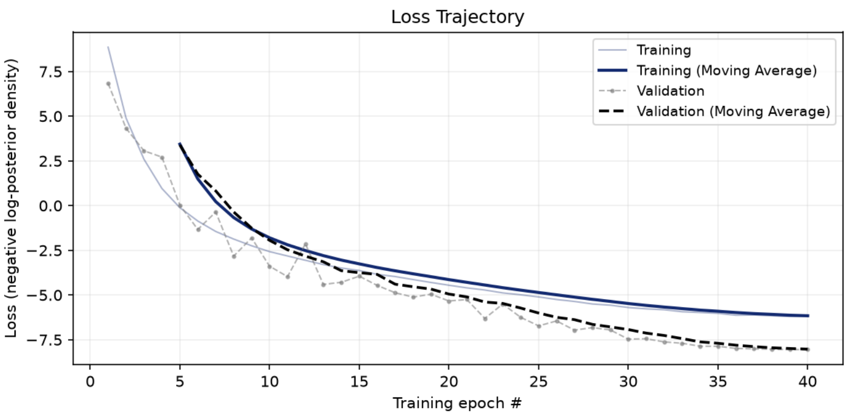

Figure 1: Loss trajectory (negative log-posterior density) of the CNN + CouplingFlow model trained on 50,000 samples: raw curves (thin) and moving averages (thick). Validation tracks training loss throughout—no overfitting.

6 Diagnostics

All diagnostics are computed on the 300 held-out test systems, with 2,000 posterior draws per system, for the final CNN + CouplingFlow model trained on 50,000 samples (both the 8-parameter and, where noted, the 9-parameter variant). We assess point accuracy (parameter recovery), uncertainty calibration (simulation-based calibration), information gain (posterior contraction and z-scores), and generative fidelity (posterior predictive checks).

6.1 Parameter Recovery

Figure 2 plots the posterior median against the ground truth for each parameter, with the posterior median absolute deviation (MAD) as the uncertainty bar and the Pearson correlation

r annotated per panel; Table 5 reports the corresponding R^2 of the posterior mean, alongside the two earlier configurations for comparison.

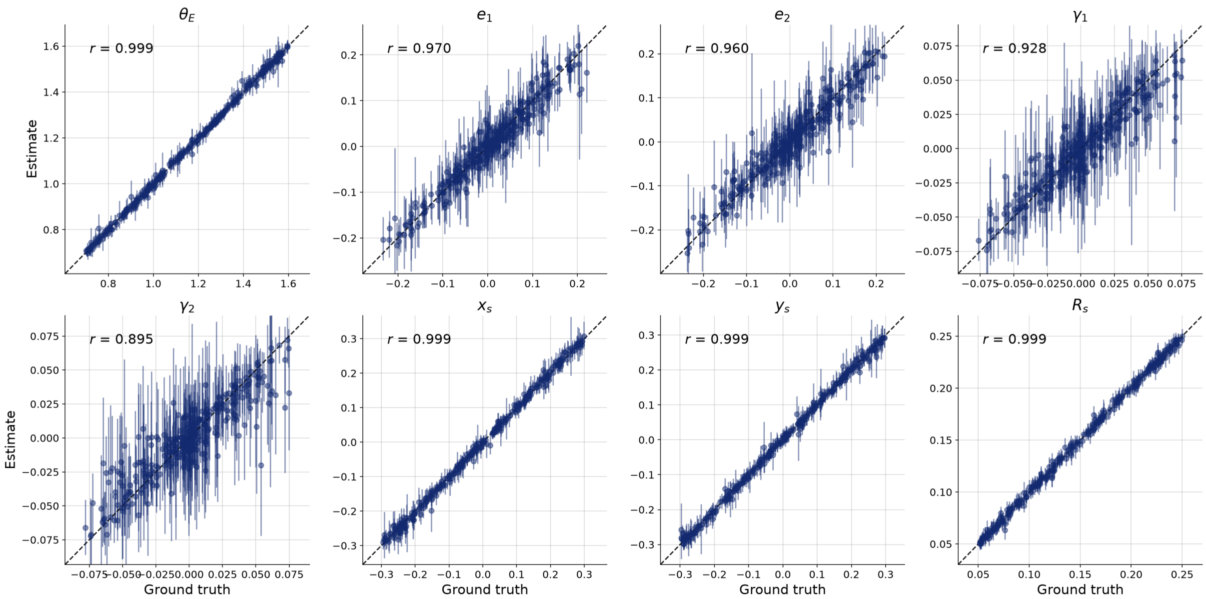

Figure 2: Ground truth vs. posterior median with MAD error bars for the 300 held-out test systems (8-parameter CNN + CouplingFlow model, 50,000 training samples). The dashed diagonal marks perfect recovery.

Table 5: Recovery accuracy (R^2, posterior mean vs. truth, 300 test systems) for the final model and the two earlier configurations of the same architecture.

Parameter	50k, SNR$\sim$16	20k, SNR$\sim$16	8k, SNR$\sim$4
θ_E	1.00	1.00	0.88
R_s	1.00	1.00	0.95
x_s	1.00	0.99	0.83
y_s	1.00	0.99	0.84
e_1	0.94	0.89	0.47
e_2	0.92	0.84	0.49
γ_1	0.86	0.80	0.21
γ_2	0.80	0.65	0.06

The recovery hierarchy follows directly from how strongly each parameter imprints on the image. The Einstein radius, source position, and source size set the ring radius, arc configuration, and arc thickness—large, direct morphological signatures—and are recovered near-perfectly (r = 0.999, 0.999, 0.999, 0.999). The lens ellipticity (r = 0.970, 0.960) imprints a quadrupole distortion on the arcs that is clearly measurable at this SNR. The external shear is the weakest signal—its modulus is at most 0.08 and its quadrupole is partially degenerate with the lens ellipticity—yet it is still meaningfully constrained (r = 0.928, 0.895), with the visibly wider MAD bars correctly reporting it as the least-identified parameter.

6.2 Simulation Based Calibration

Point accuracy is not sufficient: the posterior *width* must also be statistically correct. Simulation-based calibration (SBC) [7] exploits the self-consistency $p(\boldsymbol{\theta} \mid \mathbf{x})\, p(\mathbf{x}) = p(\mathbf{x} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})$: for

$(\boldsymbol{\theta}^{(n)}, \mathbf{x}^{(n)})$ drawn from the joint distribution, $\boldsymbol{\theta}^{(n)}$ is one draw from the posterior given $\mathbf{x}^{(n)}$, so its fractional rank among the approximator's posterior samples must be uniformly distributed if the approximation is exact. Figure 3 shows the difference between the empirical CDF of these ranks and the uniform CDF, with simultaneous 95% confidence bands [8]. The ellipticity and shear parameters stay within the bands: their posteriors show no evidence of miscalibration. The four best-constrained parameters (θ_E, x_s, y_s, R_s), in contrast, trace a symmetric S-shape that exits the band on both sides—too few extreme ranks, too many central ones. This is the signature of mildly *under*confident posteriors: the truth lands in the posterior tails less often than it should, so the credible intervals for these parameters are somewhat wider than necessary. Underconfidence is the benign direction of miscalibration—intervals are conservative rather than misleading—but it is a real, reproducible defect: it appears identically at 20,000 and 50,000 training samples (Section 6.5), which localizes its cause in the network rather than in the amount of training data.

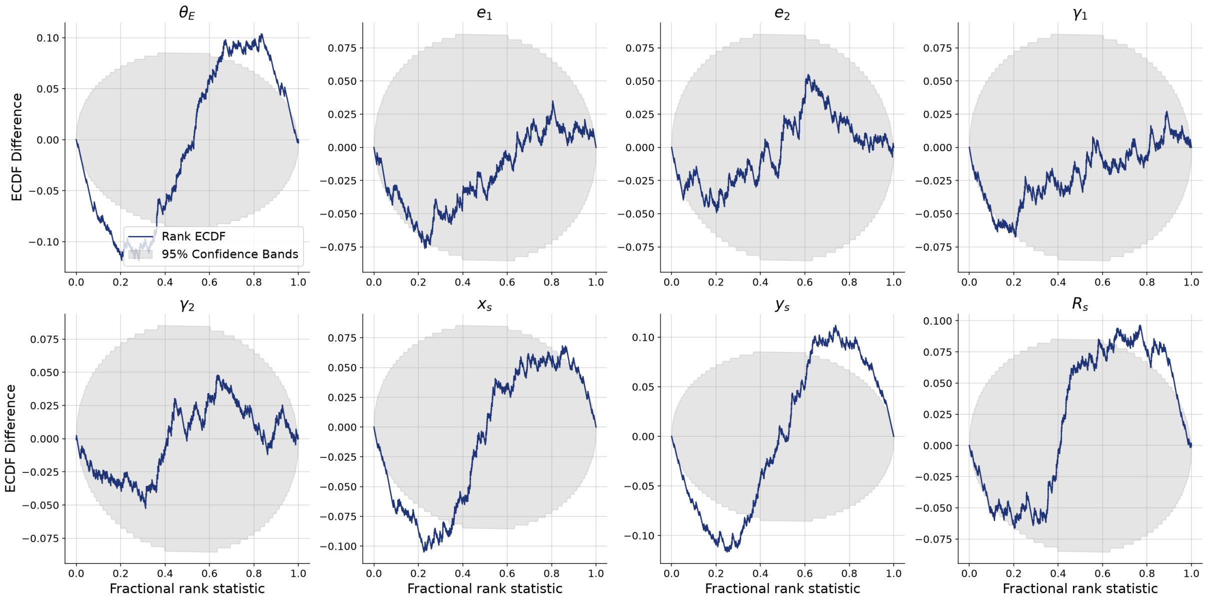

Figure 3: SBC rank-ECDF difference plots with simultaneous 95% confidence bands (8-parameter model, 50,000 training samples, 300 test systems). Curves inside the grey band are consistent with calibrated posteriors.

6.3 Posterior Contraction

Posterior contraction, $1 - \mathrm{Var}_{\mathrm{post}}/\mathrm{Var}_{\mathrm{prior}}$, quantifies how much information the image adds beyond the prior (near 1 = strong information gain), while the posterior z-score, $(\mathbb{E}[\theta \mid \mathbf{x}] - \theta^*)/\mathrm{SD}_{\mathrm{post}}$, measures the standardized bias of the estimate. Figure 4 shows the two against each other for every test system. The morphologically dominant parameters (θ_E, x_s, y_s, R_s) cluster at contraction ≈ 1 with z-scores tightly spread around 0: highly informative, unbiased posteriors. Ellipticity and shear reach intermediate contraction, exactly reflecting their weaker imprint on the data, with z-scores still centred on zero and almost entirely within ± 2—uncertain, but honest, estimates.

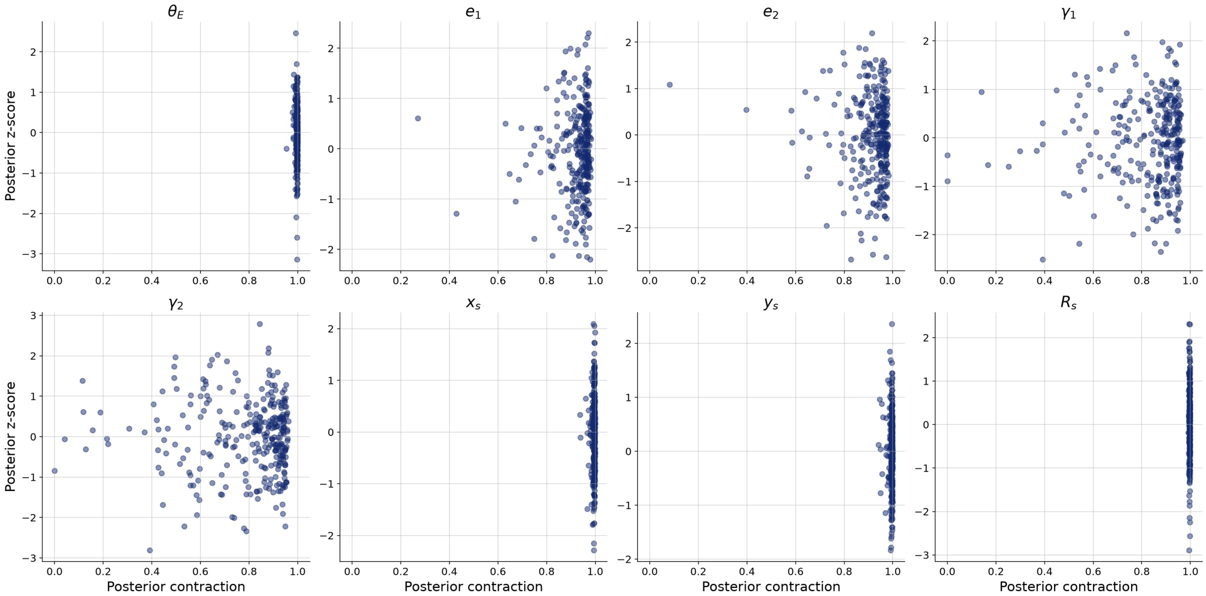

Figure 4: Posterior z-score vs. posterior contraction (8-parameter model, 50,000 training samples, 300 test systems). The ideal regime is high contraction with $|z| \lesssim 2$ spread around 0.

6.4 Posterior Predictive Check

As a complementary end-to-end test, images are re-simulated from independent posterior draws and compared with the observed image (Figure 5). The re-simulations reproduce the observed arc configuration—ring radius, knot positions, and azimuthal light distribution—showing that the posterior concentrates on parameter values that regenerate the data; residual draw-to-draw variation visualizes the remaining posterior uncertainty.

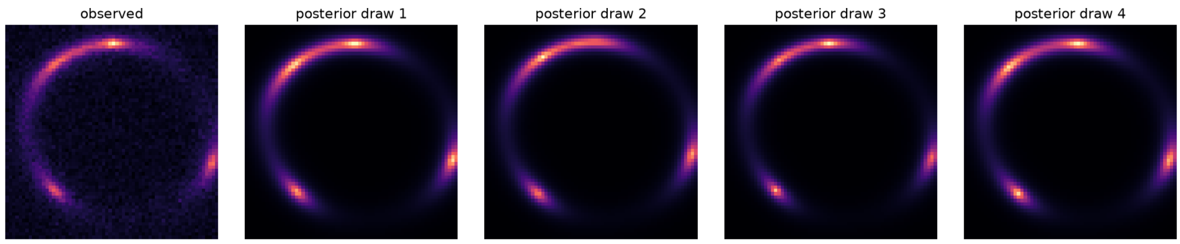

Figure 5: Posterior predictive check for one test system: observed noisy image (left) and four noise-free re-simulations from independent posterior draws.

6.5 Other Configurations

Three configurations of the same architecture were compared, differing in dataset size and SNR (diagnostic figures for the earlier configurations are in the appendix):

- 8,000 **samples, median peak SNR** ≈ 4 **(untuned source amplitude).** Validation loss plateaued by epoch $\sim$20 and then degraded while training loss kept falling (overfitting); SBC showed pronounced U-shaped rank histograms (overconfident posteriors) for θ_E, x_s, y_s, R_s; shear was essentially unrecoverable ($R^2 \leq 0.21$, Table 5).

- 20,000 **samples, median peak SNR** ≈ 16. Overfitting disappeared and recovery became excellent, but the rank-ECDF difference curves of the best-constrained parameters (θ_E, x_s, y_s, R_s) exited the 95% band in a central-tendency S-shape (Figure 9, appendix)—mild underconfidence.

- 50,000 **samples, median peak SNR** ≈ 16 **(final).** Recovery improved further, most visibly for the shear (r: $0.810 \to 0.895$ for γ_2, $0.895 \to 0.928$ for γ_1) and ellipticity components, and the validation loss dropped from -5.9 to -8.05. The mild underconfidence of the dominant parameters, however, persisted essentially unchanged (Figure 3).

The progression cleanly separates the failure modes. Raising the SNR moved the weakly-imprinting parameters (ellipticity, shear) from unrecoverable to well-recovered; enlarging the dataset repaired overfitting and sharpened recovery. That the residual underconfidence did *not* respond to a 2.5× dataset increase indicates a capacity or expressiveness limit of the current summary/inference networks rather than a data limit—the natural next knob would be a larger summary dimension or deeper flow, not more simulations.

7 Inference

7.1 Inference on a fiducial system

As the final demonstration, one "observed" image was simulated from a fixed fiducial parameter set ($\theta_E = 1.15''$, mildly elliptical lens $q \approx 0.85$, weak shear, off-centre compact source) and inference performed on it. Figure 6 compares the prior (grey) with 2,000 posterior draws (blue) in all pairwise marginals, with the truth marked. The posterior contracts sharply around the true values for every parameter, with the widest remaining spread in the shear components—consistent with every diagnostic above.

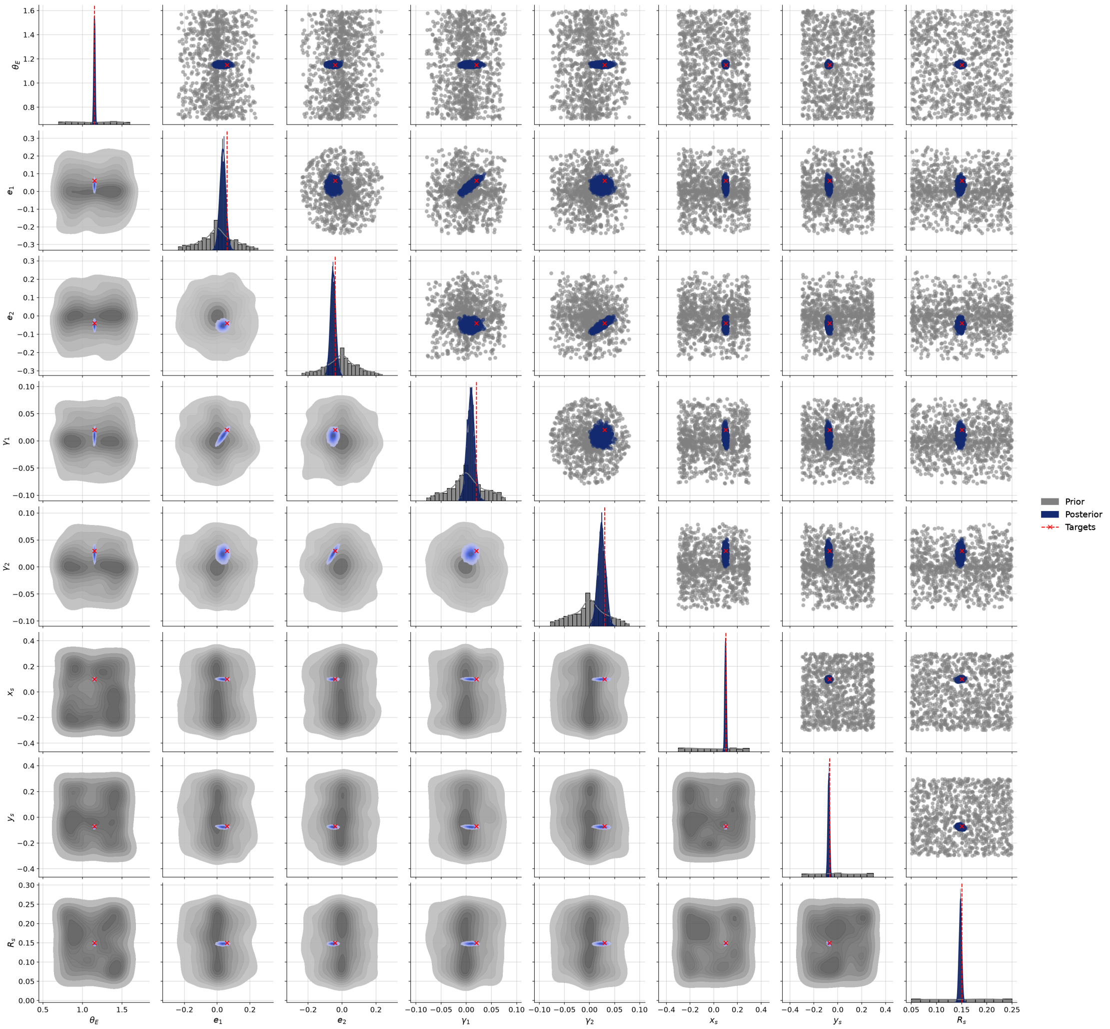

Figure 6: Posterior (blue) vs. prior (grey) for a single fiducial “observed” lens, with the true parameter values marked (red). 8-parameter CNN + CouplingFlow model trained on 50,000 samples.

7.2 External convergence and the mass-sheet degeneracy

A uniform convergence sheet κ rescales the lens equation in a way that is almost perfectly absorbed by a rescaling of the unobservable source plane [5]: image morphology alone cannot distinguish (κ, θ_E) from a suitably transformed pair. A sound inference pipeline must *report* this degeneracy rather than hide it. We repeat the full pipeline (generate, train, evaluate) with $\kappa \sim \mathcal{U}(0, 0.2)$ as a ninth inferred parameter, at identical SNR and dataset size. The results show exactly the expected signature:

- **Minimal posterior contraction.** The prior standard deviation of κ is 0.058; the mean posterior standard deviation is 0.048, a contraction of only 16%. The network has correctly learned that κ is nearly unidentifiable from one image: posterior means hover near the prior mean regardless of the truth ($R^2 = 0.21$, Figure 11, appendix), with error bars spanning almost the whole prior.
- **Strong internal anti-correlation.** Within individual posteriors, the mean correlation between κ and θ_E is -0.96: the two parameters trade off along the degeneracy direction, exactly as the lensing equations predict. This correlation is *stronger* at high SNR than

in the low-SNR pilot (-0.83): with less noise the posterior collapses onto the degeneracy ridge.

- **Degradation of θ_E only.** θ_E recovery degrades from $R^2 = 1.00$ (without κ) to 0.94 (with), while every other parameter is essentially unaffected—the unidentifiable mass sheet feeds its uncertainty specifically into the mass normalization.

That an off-the-shelf neural posterior estimator recovers this degeneracy structure without being told about it is a meaningful validation of the SBI approach: the network reports joint posterior *geometry*, not just marginal point estimates. Breaking the degeneracy requires external information (stellar kinematics, time delays, or line-of-sight priors) not present in a single image.

8 Discussion & Conclusion

We built and validated an end-to-end amortized Bayesian inference pipeline for strong-lens parameter estimation. From single noisy 64×64 images at realistic SNR, the BayesFlow posterior estimator recovers the Einstein radius, source position, and source size near-perfectly, the lens ellipticity accurately, and even the weak external shear meaningfully (Table 5); simulation-based calibration finds calibrated posteriors for ellipticity and shear and mildly conservative (underconfident) ones for the best-constrained parameters—the safe direction of miscalibration—while posterior contraction confirms strong, unbiased information gain for all morphologically dominant parameters. When external convergence is included, the inferred posteriors exhibit the mass-sheet degeneracy exactly as theory predicts: near-zero contraction of κ, a strong κ–θ_E anti-correlation (-0.96), and a corresponding, isolated degradation of θ_E recovery.

The project's main practical constraint was the cost of simulation and training, which shaped the workflow: offline, CPU-parallelized data generation; GPU training (a 4–5$\times$ per-step speedup over CPU made the 50,000-sample configuration feasible); and a stepped dataset-size study (8k $\rightarrow$ 20k $\rightarrow$ 50k) in which each increase fixed a specific, diagnosable defect—first overfitting, then residual SBC miscalibration. The forward model still fixes several quantities (lens centroid, source amplitude and profile, PSF) that are uncertain for real observations and omits the lens-galaxy light; extending the parameter vector with these nuisances within the same pipeline is the natural next step toward survey data. Once trained, amortized posterior sampling costs milliseconds per system—the property that makes this approach attractive for the 10^4–10^5 lenses expected from upcoming surveys.

References

[1] S. T. Radev, U. K. Mertens, A. Voss, L. Ardizzone, and U. Köthe, "BayesFlow: Learning complex stochastic models with invertible neural networks," *IEEE Transactions on Neural Networks and Learning Systems*, 33(4):1452–1466, 2020. https://doi.org/10.1109/TNNLS.2020.3042395

[2] S. T. Radev, M. Schmitt, L. Schumacher, L. Elsemüller, V. Pratz, Y. Schälte, U. Köthe, and P.-C. Bürkner, "BayesFlow: Amortized Bayesian workflows with neural networks," *Journal of Open Source Software*, 8(89):5702, 2023. https://doi.org/10.21105/joss.05702

[3] S. Birrer and A. Amara, "lenstronomy: Multi-purpose gravitational lens modelling software package," *Physics of the Dark Universe*, 22:189–201, 2018. https://doi.org/10.1016/j.dark.2018.11.002

[4] R. Kormann, P. Schneider, and M. Bartelmann, "Isothermal elliptical gravitational lens models," *Astronomy & Astrophysics*, 284:285–299, 1994.

[5] E. E. Falco, M. V. Gorenstein, and I. I. Shapiro, "On model-dependent bounds on H_0 from gravitational images: Application to Q0957+561A,B," *The Astrophysical Journal*, 289:L1–L4, 1985. https://doi.org/10.1086/184422

[6] P. Schneider and D. Sluse, "Mass-sheet degeneracy, power-law models and external convergence: Impact on the determination of the Hubble constant from gravitational lensing," *Astronomy & Astrophysics*, 559:A37, 2013. https://doi.org/10.1051/0004-6361/201321882

[7] S. Talts, M. Betancourt, D. Simpson, A. Vehtari, and A. Gelman, "Validating Bayesian inference algorithms with simulation-based calibration," *arXiv:1804.06788*, 2018.

[8] T. Säilynoja, P.-C. Bürkner, and A. Vehtari, "Graphical test for discrete uniformity and its applications in goodness-of-fit evaluation and multiple sample comparison," *Statistics and Computing*, 32:32, 2022. https://doi.org/10.1007/s11222-022-10090-6

[9] C. R. Harris, K. J. Millman, S. J. van der Walt, et al., "Array programming with NumPy," *Nature*, 585:357–362, 2020. https://doi.org/10.1038/s41586-020-2649-2

Appendix: Additional Figures

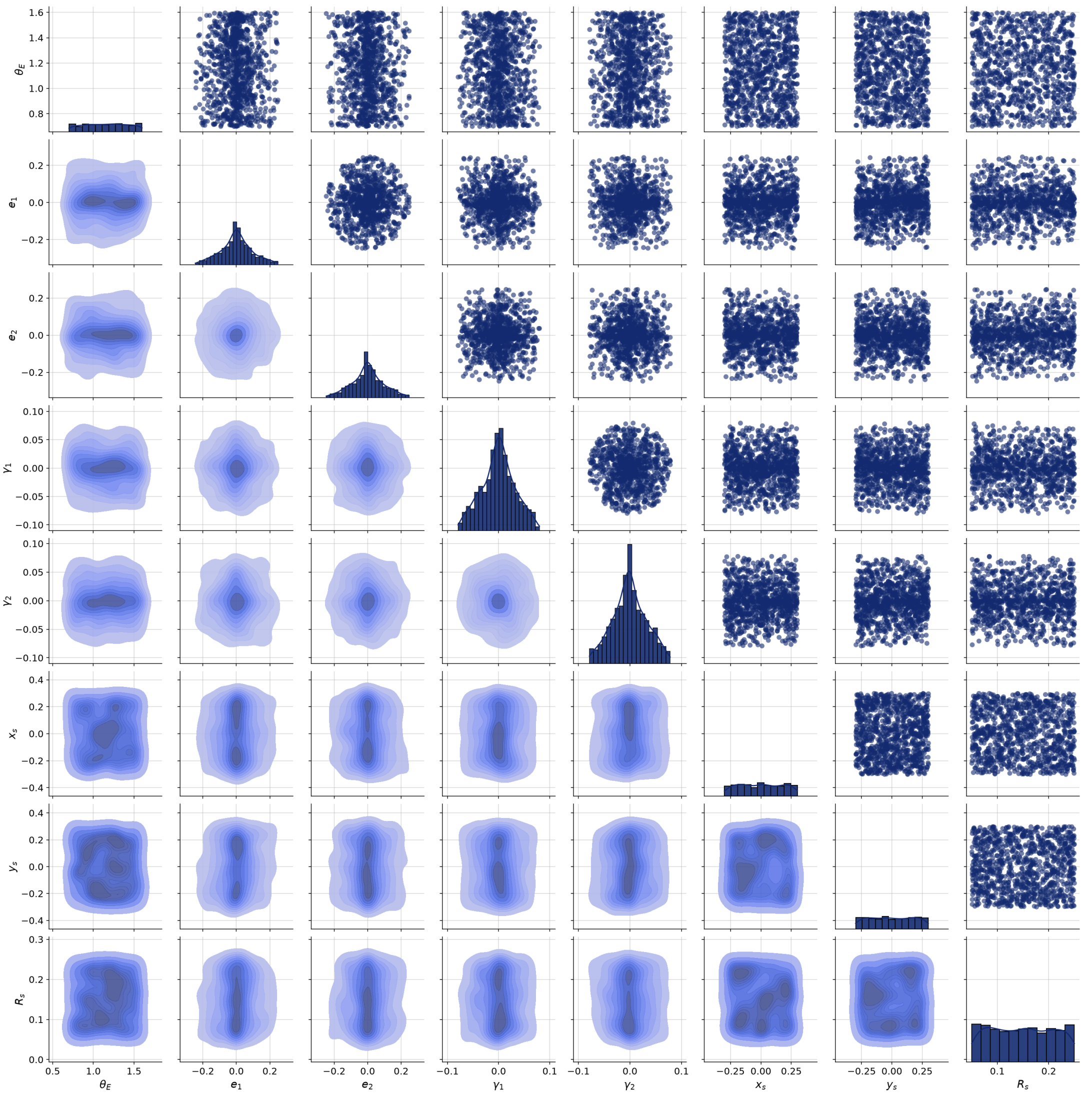

Figure 7: Marginal and pairwise joint distributions of prior draws from the training set. The ring-shaped (e_1, e_2) and (γ_1, γ_2) supports are induced by sampling in (axis ratio, angle) and (modulus, angle) respectively and converting to Cartesian components.

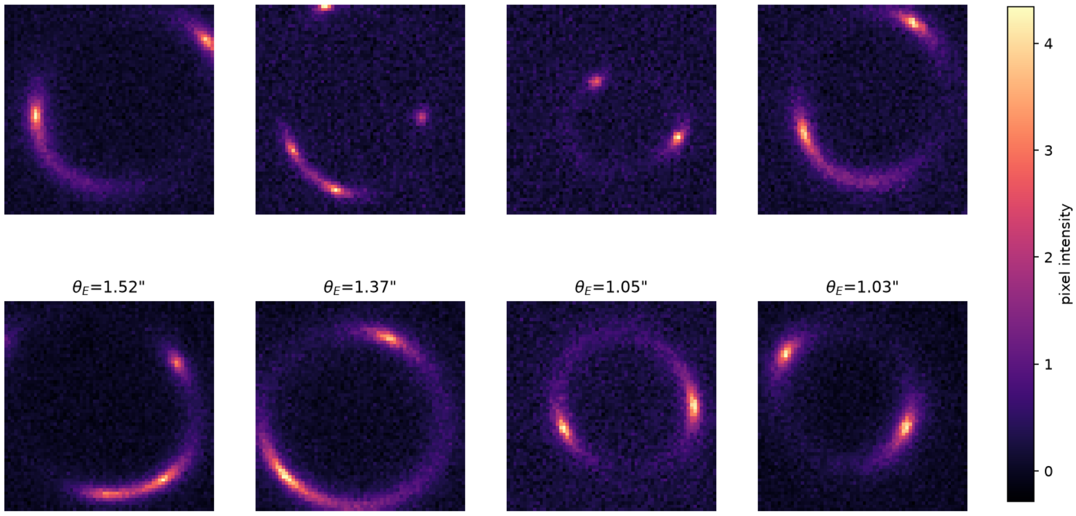

Figure 8: Eight example noisy images from the training set (median peak SNR ≈ 16), with the true Einstein radius of each system.

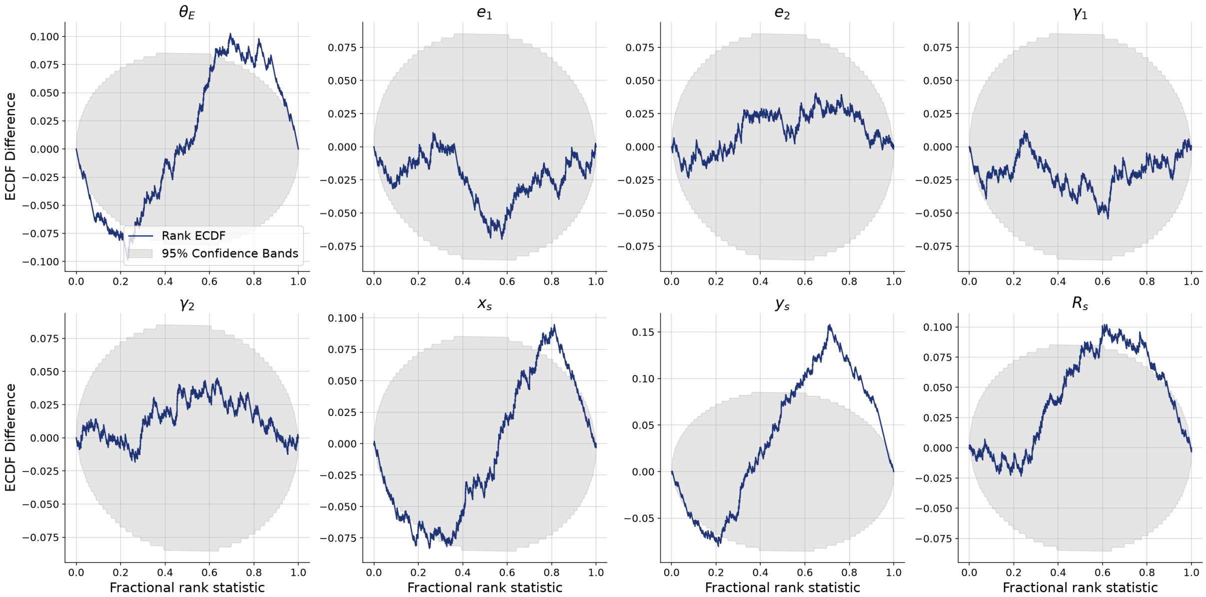

Figure 9: SBC rank-ECDF difference plots for the intermediate configuration (20,000 training samples, same architecture and SNR as the final model). The best-constrained parameters (θ_E, x_s, y_s, R_s) exit the 95% band—the residual miscalibration that motivated the increase to 50,000 samples (compare Figure 3).

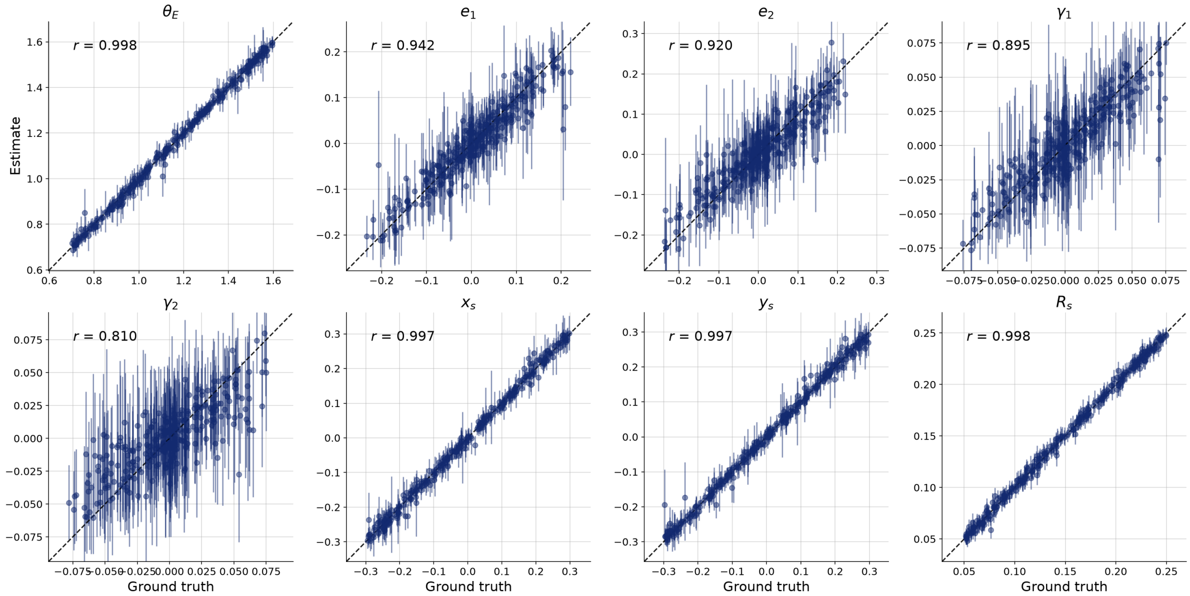

Figure 10: Parameter recovery for the intermediate 20,000-sample configuration (posterior median, MAD error bars, Pearson r; same style as Figure 2).

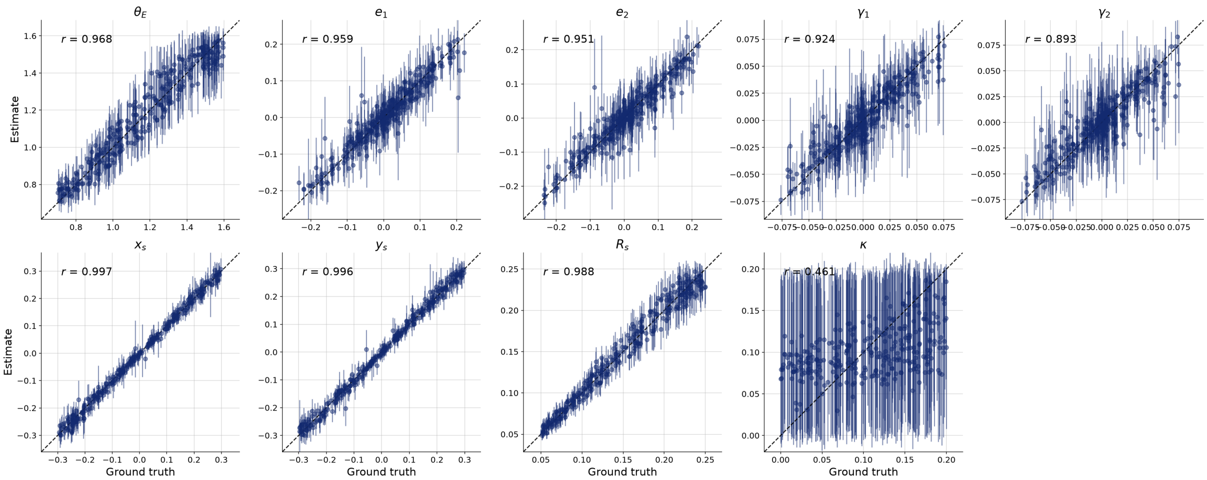

Figure 11: Parameter recovery for the 9-parameter model including external convergence (50,000 training samples). The κ posterior stays near the prior mean with near-prior-width uncertainties, and the θ_E panel shows the degeneracy-induced widening relative to Figure 2.

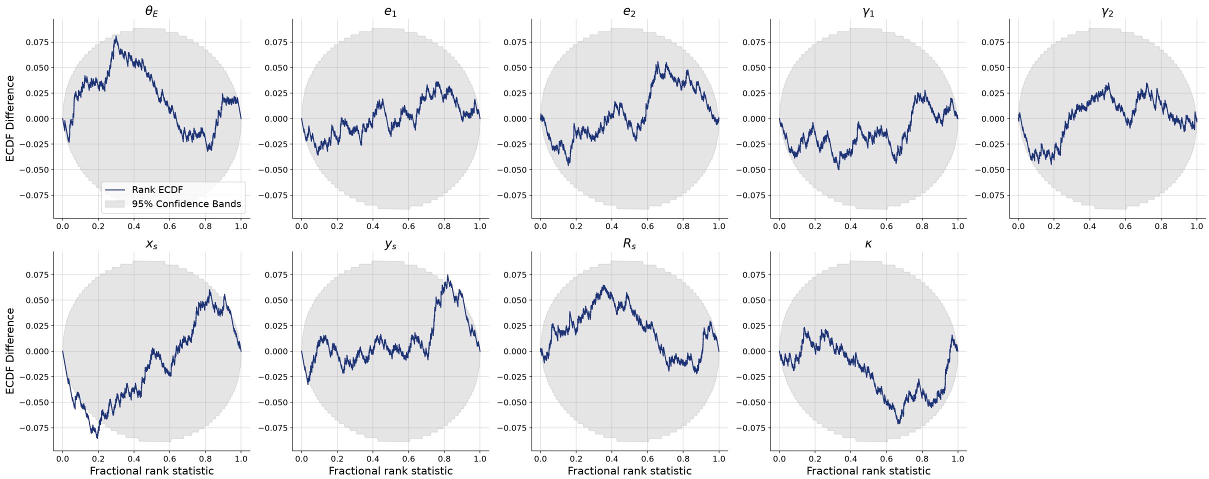

Figure 12: SBC rank-ECDF difference plots for the 9-parameter (external convergence) model at 50,000 training samples.

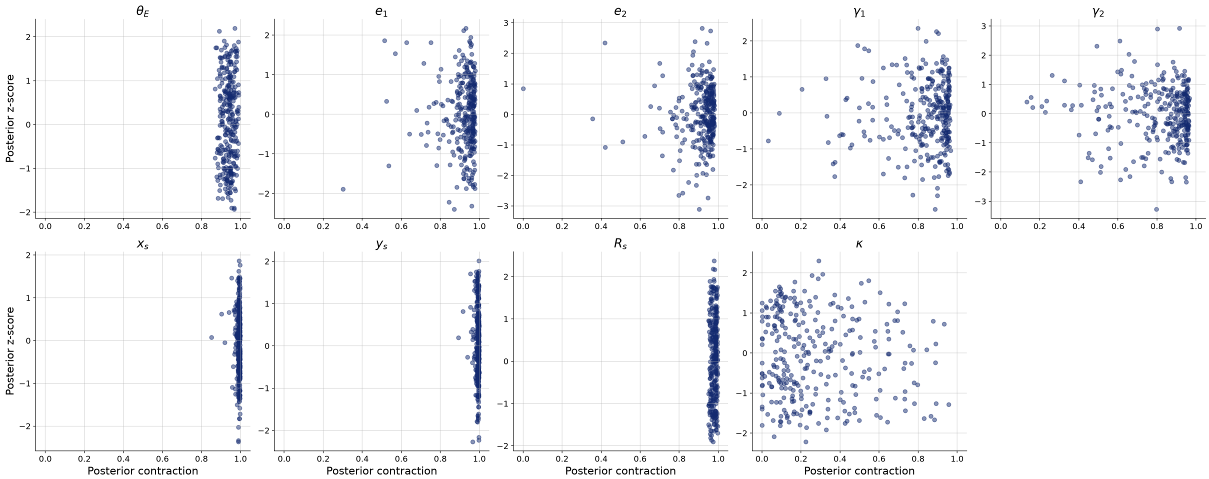

Figure 13: Posterior z-score vs. contraction for the 9-parameter model. The κ panel sits at near-zero contraction—the mass-sheet degeneracy—while all other parameters match the 8-parameter model (Figure 4).

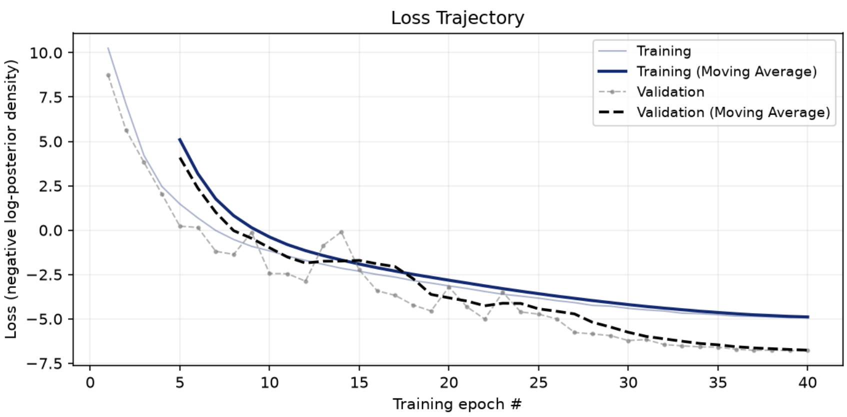

Figure 14: Loss trajectory of the 9-parameter (external convergence) model, 50,000 training samples.

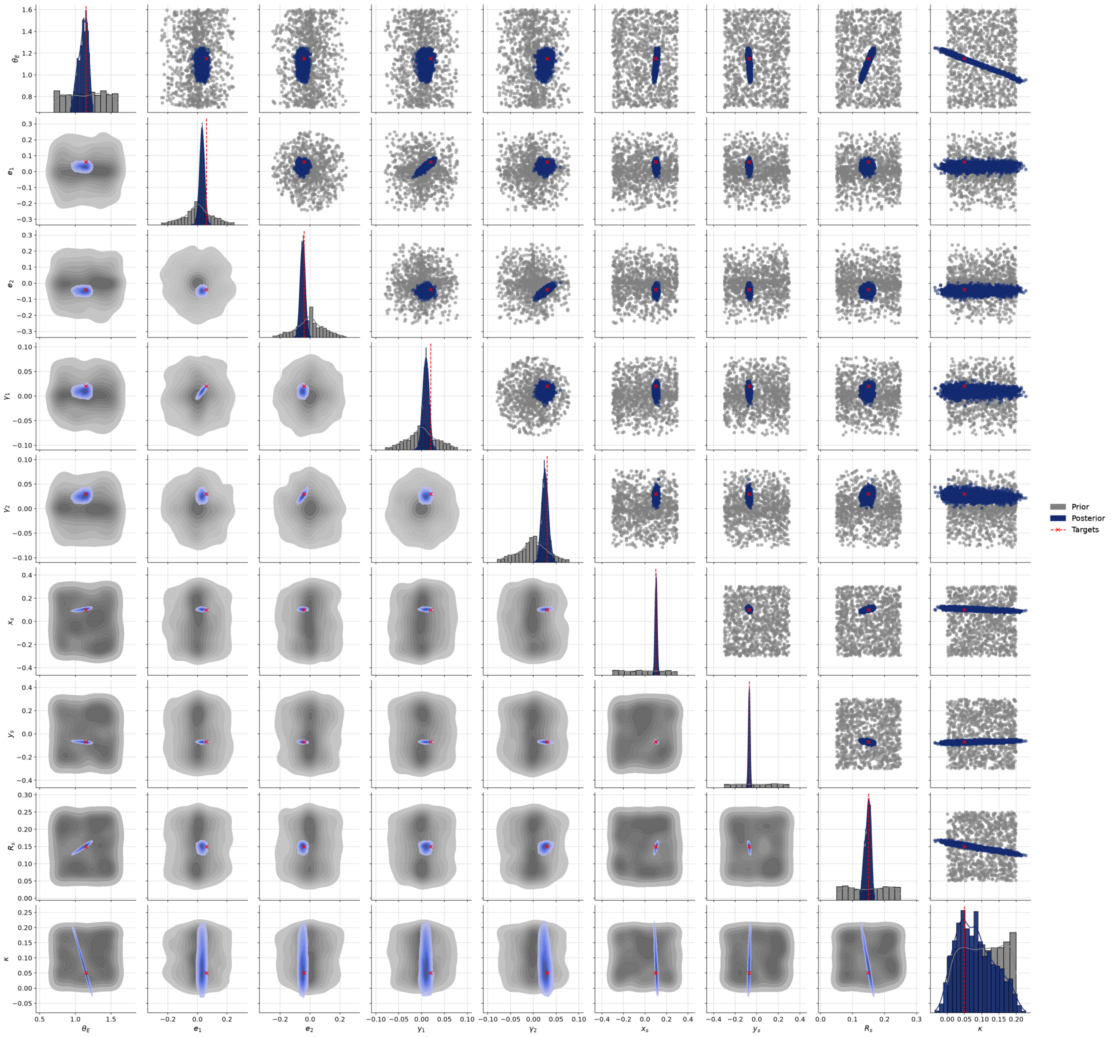

Figure 15: Posterior vs. prior for the fiducial system under the 9-parameter model. The κ marginal barely contracts from its prior, and the (κ, θ_E) panel shows the negative degeneracy ridge.